Maths Exploring & Assessment at Terminal-06

MCQ A · diversion and convexity

After S29; detrievable function over each interval considered.

For each question, only one proposal is accurate.
consult the correction.

1. If $f''(x) > 0$ over an interval, f is:

A. Decreasing
B. Convex
C. Concave
D. constant

2. The derivative of e^{1+x^2} is:

A. e^{1+x^2}
B. $2xe^{x}$
C. $2xe^{1+x^2}$
D. $1 + 2x$

3. A tangent to a convex function is:

A. Always above
B. still below the interval
C. parallel to the axis
D. None

4. If $f'(x)$ change of sign of $+$ to $-$ in a, then f :

A. has a minimum in a
B. has a local maximum in a
C. is convex
D. is not continuous

5. The function $x \mapsto |x|$ is:

A. Derivable in 0
B. Not continuous in 0
C. continuous but not driftable in 0
D. Concave on $\mathbb{R}$

MCQ B · Continuity and VIT

After S33.

For each question, only one proposal is accurate.
consult the correction.

6. To apply the VIT to $f(x) = 0$ on $[a; b]$, we check first:

A. that f is continuous on $[a; b]$

B. That f' is null

C. That $a = b$

D. That f is affine

7. If $f(a) < 0 < f(b)$ and f is continuous on $[a; b]$, then:

A. f has exactly one root

B. f has at least one root

C. f is increasing

D. $f(a) = f(b)$

8. To demonstrate the uniqueness of a solution, it is sufficient, for example, that f either:

A. Limited

B. Strictly monotone

C. pair

D. Derivable from a only

9. The dichotomy method maintains at each stage:

A. two bounds of the same sign

B. an interval containing a root with signs opposing the bounds

C. a horizontal tangent

D. an exact root value

10. A curve of f'' negative indicates that f' is:

A. increasing

B. Decreasing

C. Positive

D. None

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Any continuous function allows a root.

2. If f is convex, so f' is increasing where it is defined.

3. An approximate value obtained by dichotomy is evidence of existence.

Review review

I know how to associate sign of f'', convexity and variations in f' ; isolate and then justify a solution. **To be monitored:** announce uniqueness without monotonicity; confuse a graphic conjecture with the VIT.